

Executive Data Analysis Report

Project Identifier: OLIST-LATE-DELIVERY-RISK

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1. Executive Summary

The model's OVERALL ACCURACY of 74.9% is below the majority-class baseline accuracy of 91.8% when predicting whether an order will be a late delivery, meaning a naive model that always predicts on-time delivery would score higher for orders in this dataset. This is the expected tradeoff of correcting for class imbalance, as the model sacrifices raw accuracy to catch more actual cases of late delivery, which an always-guess-on-time baseline for orders never attempts. The model's ability to rank and separate orders into those at risk of late delivery versus those on time yields a ROC-AUC of 0.781. At the default threshold of 0.5, the model correctly identifies 1,043 orders as cases of late delivery and 13,397 orders as on-time delivery. However, the model misses 538 orders that were actually instances of late delivery and falsely flags 4,291 orders as late delivery when they were actually on-time.

2. Data Diagnostics Profile

Input shape	96,476 rows × 22 columns
Output shape	96,476 rows × 22 columns
Duplicate rows removed	0
High-missing columns flagged	None
Low-variance columns dropped	None
Columns type-corrected	None
Numeric columns imputed	customer_zip_code_prefix, n_items, n_distinct_sellers, n_distinct_products, total_price, total_freight, avg_product_weight_g, avg_product_photos_qty, total_payment_value, max_installments, n_payment_methods, purchase_to_estimated_days, is_late_delivery
Categorical columns imputed	order_id, order_purchase_timestamp, order_estimated_delivery_date, customer_unique_id, customer_city, customer_state, primary_seller_state, product_category, primary_payment_type

3. Model Performance Leaderboard

Task type: binary_classification | **Best model:** HistGradientBoostingClassifier

Candidate Model	CV Score
HistGradientBoostingClassifier	0.5693
RandomForestClassifier	0.5632
LogisticRegression	0.4738

F1 (macro)	0.5745
Precision (macro)	0.5785
Recall (macro)	0.7086
ROC-AUC	0.7807

	Predicted Negative (0)	Predicted Positive (1)
Actual Negative (0)	13,397	4,291
Actual Positive (1)	538	1,043

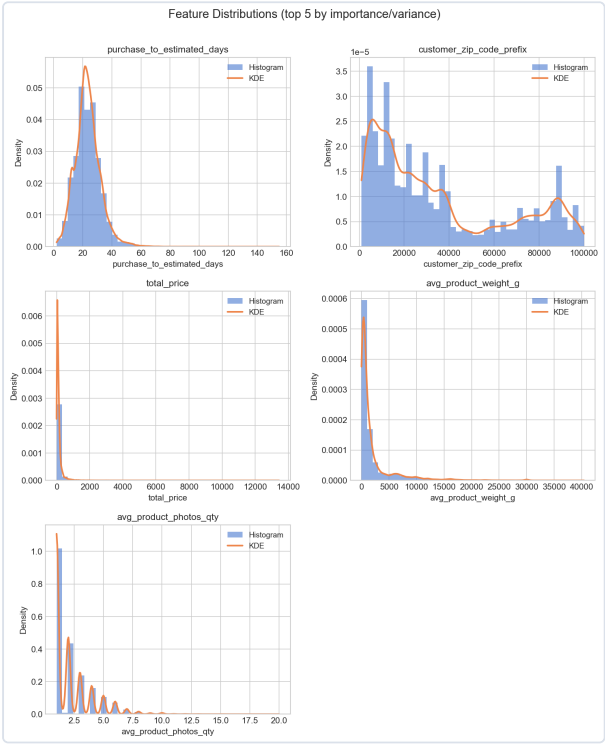
Threshold	Precision	Recall	F1 (macro)
0.3	0.122	0.908	0.399
0.4	0.154	0.796	0.503
0.5	0.196	0.660	0.574

Note on probabilities: this model uses `class_weight='balanced'` to handle class imbalance, which systematically shifts its raw predicted probabilities away from true likelihoods. Treat each decision threshold above (e.g. `threshold=0.3`) as a precision/recall cutoff chosen for a business tradeoff, not a literal percentage chance of the outcome. Measured calibration error (ECE) for this model: 0.306.

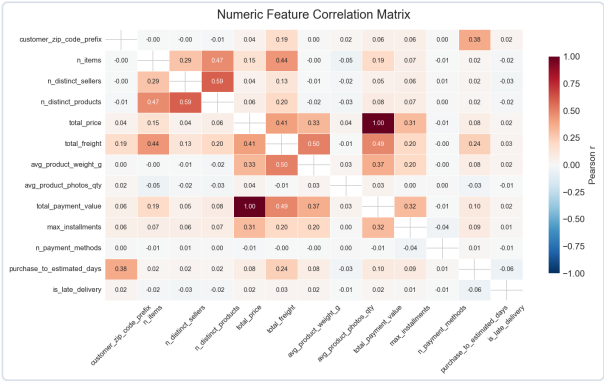
This model uses `class_weight='balanced'` to handle class imbalance, which systematically shifts `predict_proba` outputs away from true probabilities -- exactly what a calibration curve measures. Treat a decision threshold (e.g. `threshold=0.3` in the threshold sweep) as a precision/recall cutoff chosen for a business tradeoff, not a literal percentage chance of the outcome. Measured ECE = 0.3061 exceeded the 0.05 threshold, so a isotonic-calibrated comparison model was fit ('HistGradientBoostingClassifier' + CalibratedClassifierCV, `cv=3`). Its ECE = 0.0070 vs. the raw model's 0.3061 (lower is better). This comparison is diagnostic only -- the serialized production model is still the raw, uncalibrated estimator.

4. Automated Business Insights

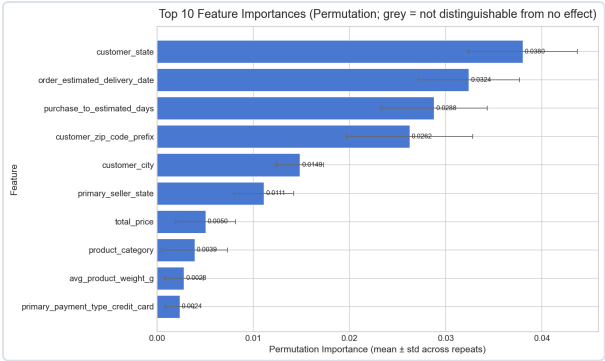
The customer's state of residence is associated with cases of late delivery in orders, capturing regional differences in shipping logistics. The estimated delivery date set for an order is linked to cases of late delivery in orders, reflecting the promised arrival timeline. The number of days between purchase and the estimated delivery date is linked to late delivery instances in orders, representing the total promised transit window. The customer's zip code prefix is associated with late delivery in orders, mapping local delivery zone logistics. The customer's city is linked to cases of late delivery in orders, pinpointing metropolitan or regional logistics differences. The state of the primary seller is associated with late delivery in orders, representing the inventory's geographic origin. In the held-out test set, false-positive rates are elevated across all customer segments, meaning the model frequently misclassifies on-time orders as cases of late delivery.



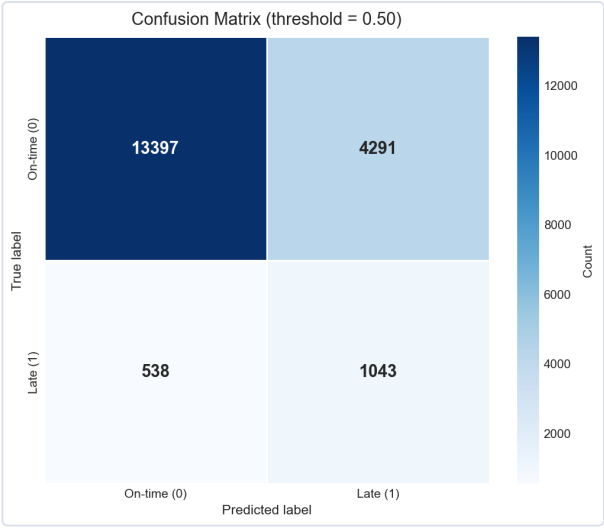
Distributions



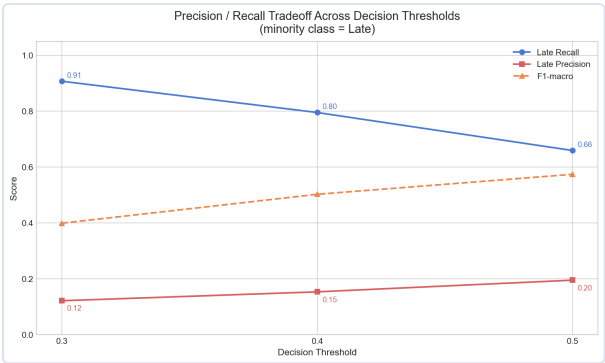
Correlation Heatmap



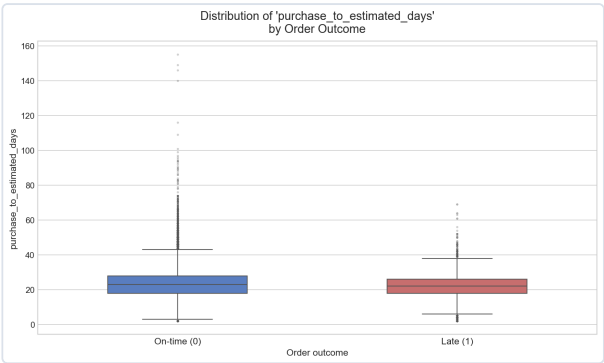
Feature Importance



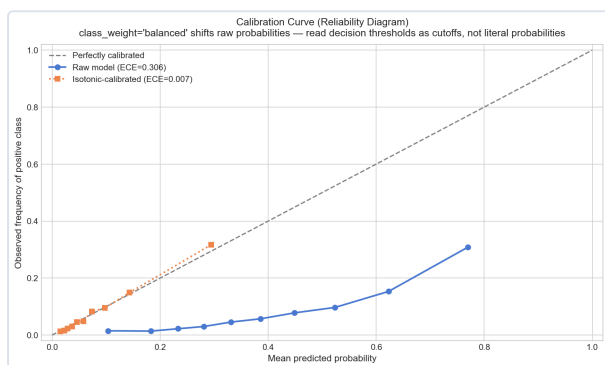
Confusion Matrix



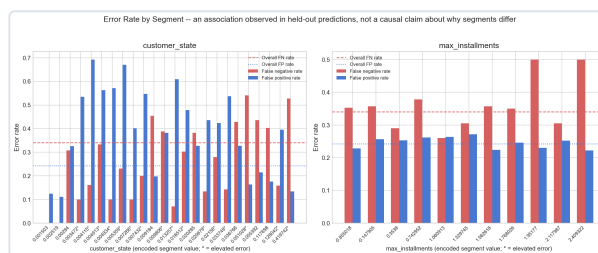
Threshold Tradeoff



Top Feature Vs Target



Calibration Curve



Error By Segment

5. Strategic Recommendations

Set the classification threshold to 0.3 for proactive customer service intervention, aiming to catch as many cases of late delivery as possible. At this threshold, the model achieves a recall of 0.908, catching 90.8% of actual orders that are instances of late delivery, though its precision is only 0.122, meaning many flagged orders will require manual verification before taking action. Set the classification threshold to 0.4 for a balanced approach that manages case volumes while maintaining reasonable catch rates. At this threshold, the model yields a precision of 0.154 and a recall of 0.796, ensuring that over 79% of actual cases of late delivery are caught while keeping the manual review workload manageable. Set the classification threshold to 0.5 for a stricter verification process that focuses resources on the highest-confidence alerts regarding late delivery. At this threshold, the model provides a precision of 0.196, meaning roughly 1 in 5 flagged orders is an actual case of late delivery, paired with a recall of 0.660.